Week 1: Introduction to AI & ML + Project 1 (Diabetes Prediction Model)

Topics Covered:

- What is Artificial Intelligence (AI)? (With daily life examples: Google Maps, YouTube recommendations, Alexa, etc.)
- What is Machine Learning (ML)?
- Types of ML:
    1. **Supervised Learning** (predict exam marks based on study hours)
    2. **Unsupervised Learning** (grouping friends by hobbies)
    3. **Reinforcement Learning** (training a dog with rewards)
- Basic ML workflow:
  Data → Cleaning → Splitting → Training → Testing → Prediction

Hands-on Project: Diabetes Prediction Model

- Dataset: Diabetes dataset (patient details like glucose level, age, BMI).
- Steps:
    1. Import dataset
    2. Clean the data (handle missing values)
    3. Train ML model (Logistic Regression)
    4. Predict whether a person has diabetes or not

🌟 1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) means making machines or computers **think and act like humans**.

👉 Example in real life:

- **Google Maps**: Suggests shortest path just like your friend guiding you.
- **YouTube Recommendations**: Shows videos you like, as if it "knows" your taste.
- **Alexa / Siri**: They understand your voice and give answers, just like a human helper.
- **Self-driving Cars**: The car "sees" the road and drives safely without a driver.

So, **AI = Human-like intelligence in machines.**

🌟 2. What is Machine Learning (ML)?

Machine Learning is a **part of AI**.
It simply means: **teaching machines to learn from data and improve by themselves** without being directly programmed.

👉 Example:

- When you use **Netflix**, it learns what type of movies you like (horror, comedy, romantic). Next time, it automatically suggests similar movies.
- When you post a photo on **Facebook/Instagram**, it recognizes your face and tags you.

So, **ML = Machine learns from past data to predict future outcomes.**

🌟 3. Types of Machine Learning

There are mainly **3 types of ML**. Let's understand with simple examples:

(a) Supervised Learning – Learning with Answers

- In this method, the machine learns from **labeled data** (data with correct answers already given).
- Example: If you study "hours studied" vs "marks scored" data, you can predict your future marks.

📌 Student Example:
If a teacher gives you many math problems with solutions, you learn faster. Similarly, the machine learns from solved examples.

(b) Unsupervised Learning – Learning without Answers

- Here, the machine is given **raw data** (without answers). It tries to find hidden patterns.
- Example: Grouping students in a class based on hobbies (sports lovers, gamers, music lovers).

📌 Student Example:
Imagine you are new in college, and you don't know anyone. You try to find your group by observing who likes cricket, who likes music, etc. That's clustering.

(c) Reinforcement Learning – Learning by Trial & Error

- Here, the machine learns by **reward and punishment**, just like humans.
- Example: Training a dog. If it sits when you say "Sit!", you give it a biscuit (reward). If not, no reward.

📌 Student Example:
Think of playing a video game. At first, you lose many times, but you slowly learn what actions give you high scores. That’s reinforcement learning.

🌟 4. Basic Machine Learning Workflow

ML is like cooking a dish step by step. The steps are:

1. **Collect Data** → Like collecting ingredients for food.
   (Example: Patients' health data for diabetes.)

2. **Clean Data** → Remove bad or missing values.
   (Example: If someone forgot to write their age, we handle that.)

3. **Split Data** → Divide into 2 parts:

   - Training Data (used for teaching machine)

   - Testing Data (used for checking machine's performance)

4. 👉 Example: You study from 80% of syllabus (training) and test yourself with 20% (testing).

5. **Train the Model** → Feed training data to ML algorithm.
   (Machine learns patterns.)

6. **Test the Model** → Check accuracy using test data.

7. **Prediction** → Use model for new cases.
   (Example: Predict if a new patient has diabetes or not.)

🌟 5. Hands-on Project: Diabetes Prediction Model

📌 Problem:

We want to make a model that can **predict if a person has diabetes** based on their health details (glucose level, BMI, age, etc.).

📊 Dataset:

- Contains information like:
    - Age
    - Glucose Level
    - BMI (Body Mass Index)
    - Blood Pressure
    - Insulin level
    - …and whether the person had diabetes (Yes/No).

🔧 Steps to Build the Model:

1. **Import Dataset**
    - Load the diabetes dataset into Python.
2. **Clean the Data**
    - Check for missing or wrong values.
    - Example: If a patient's age is missing, we can fill it with an average value.
3. **Train the Model**
    - Use **Logistic Regression Algorithm**.

- Logistic Regression is a simple algorithm that answers **Yes/No** questions.
  👉 Example: Just like a doctor asks "Do you have high sugar? Do you have high BMI?" and then predicts diabetes.

4. **Test the Model**

   - Use test data to check accuracy.

   - Example: If model says "Yes – Diabetes" for 8 out of 10 correct cases, accuracy is 80%.

5. **Make Prediction**

   - Now, give new patient data and let model predict.
     👉 Example: Patient (Age: 45, Glucose: 180, BMI: 28) → Model says "Diabetes: Yes".

🌟 6. Real-Life Analogy for Students

👉 Imagine a **doctor**:

- He has seen 1000 patients in the past.
- He remembers their symptoms and results.
- When a new patient comes, he compares symptoms with past cases and gives prediction.

Our **Diabetes ML Model** works exactly like this doctor.

✅ By the end of Week 1:

- You will understand **AI, ML basics, types, workflow**.
- You will complete your **first project – Diabetes Prediction Model**.